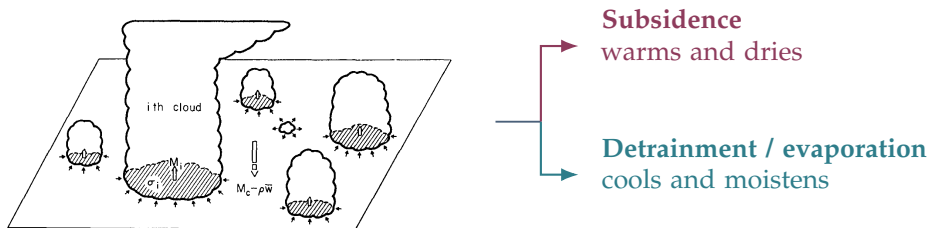


Arakawa–Schubert (1974): cloud feedbacks

1



$$M_c = \sum_i \rho \sigma_i w_i$$

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + \text{LS}_s$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + \text{LS}_q$$

Mass flux and detrainment set the environmental heating and moistening

Cloud type λ : fractional entrainment.

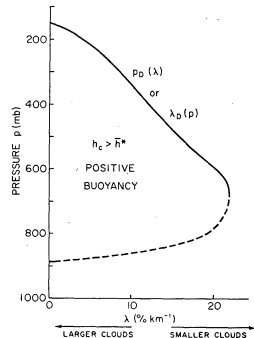
$$\partial_z \eta = \lambda \eta \quad \Rightarrow \quad \eta = e^{\lambda(z-z_B)}$$

$$\eta h_c = h_M + \lambda \int_{z_B}^z \eta \bar{h} dz'$$

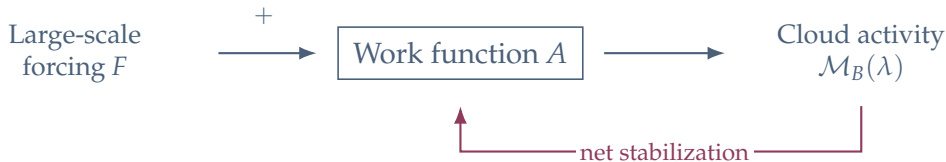
h : moist static energy. h_M : mixed-layer value.

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

$$h_c(z_D) = \hat{h}^* \approx \bar{h}^*$$



More entrainment gives a lower cloud top in the reference sounding



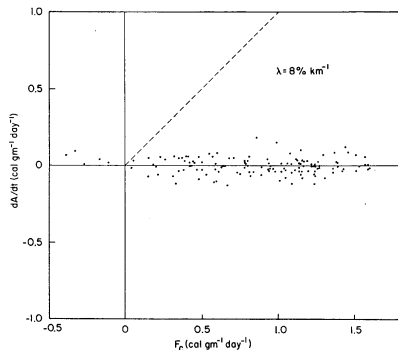
$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \eta(z, \lambda) \underbrace{\frac{g [s_{vc}(z, \lambda) - \bar{s}_v(z)]}{c_p \bar{T}(z)}}_{\text{buoyancy}} dz$$

$$\frac{dA}{dt} = \int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0 \qquad \mathcal{M}_B(\lambda) \geq 0$$

Fast cloud adjustment balances the large-scale forcing

Observations support the adjustment balance

4



Marshall Islands. Dashed line: $dA/dt = F_C$.

Timescale separation

$$\tau_{\text{adj}} \sim 10^3 - 10^4 \text{ s}$$

$$\tau_{\text{LS}} \gtrsim 10^5 \text{ s}$$

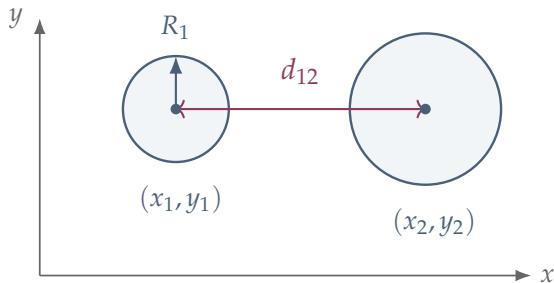
$$\text{Observed: } |dA/dt| \ll |F_C|$$

Clouds nearly balance the forcing.

The observed work function changes much less than the forcing alone predicts

Formulated: mass flux by cloud type, $\mathcal{M}_B(\lambda)$.

Spatial quantities (schematic)



Not reported for thermals

Size distribution R_i

Positions (x_i, y_i)

Distances d_{ij}

A mass-flux spectrum leaves the horizontal thermal field unspecified